

3NF Synthesis (RM3)

1. Minimal Cover

rest_phone $\rightarrow$ rest_name

rest_phone $\rightarrow$ street

rest_phone $\rightarrow$ zip

rest_phone $\rightarrow$ city

rest_phone $\rightarrow$ state

rest_phone $\rightarrow$ cuisine

rest_phone $\rightarrow$ price_range

rest_phone $\rightarrow$ image_url

rest_phone $\rightarrow$ rest_created

(rest_phone, slot_date, slot_time) $\rightarrow$ slot_capacity

(rest_phone, day_of_week) $\rightarrow$ open_time

(rest_phone, day_of_week) $\rightarrow$ close_time

(rest_phone, day_of_week) $\rightarrow$ is_closed

email $\rightarrow$ pwd_hash

email $\rightarrow$ user_role

email $\rightarrow$ fname

email $\rightarrow$ lname

email $\rightarrow$ user_phone

email $\rightarrow$ acct_created

email $\rightarrow$ employer_phone

res_num $\rightarrow$ party_size

res_num $\rightarrow$ special_req

res_num $\rightarrow$ res_status

res_num $\rightarrow$ res_created

res_num $\rightarrow$ slot_date

res_num $\rightarrow$ slot_time

res_num $\rightarrow$ rest_phone

res_num $\rightarrow$ email

(email, rest_phone) $\rightarrow$ rating

(email, rest_phone) $\rightarrow$ comment

(email, rest_phone) $\rightarrow$ review_created

(res_num, changed_at) $\rightarrow$ changed_to

(res_num, changed_at) $\rightarrow$ changed_by

Step 2: Group Attributes by Determinant

rest_phone →

{

rest_name,
street,
zip,
city,
state,
cuisine,
price_range,
image_url,
rest_created

}

(rest_phone, slot_date, slot_time) →

{

slot_capacity

}

(rest_phone, day_of_week) →

{

open_time,
close_time,
is_closed

}

email →

pwd_hash,
user_role,
fname,
lname,
user_phone,
rest_created,
employer_phone

}

res_num →

{

party_size,
special_req,
res_status,
res_created,
slot_date,
slot_time,
rest_phone,
email

}

(email, rest_phone) →

{

rating,
comment,
review_created

}

(res_num, changed_at) →

{

changed_to
changed_by

}

Step 3: Create Tables

Restaurant (

rest_phone PK,
rest_name,
street,
zip,
city,
state,
wine,
price_range,
image_url,
rest_created
)

User Account (

email PK,
pwd_hash,
user_role,
fname,
lname,
user_phone,
acct_created,
employer_phone FK
)

ReservationSlot (

rest_phone PK/FK,
slot_date PK,
slot_time PK,
slot_capacity
)

Reservation (

res_num PK
party_size,
special_req,
res_status,
res_created,
slot_date,
slot_time,
rest_phone FK,
email FK
)

RestaurantHours (

rest_phone PK/FK
day_of_week PK
open_time,
close_time,
is_closed
)

Review (

email PK/FK,
rest_phone PK/FK,
rating,
comment,
review_created
)

ReservationHistory (

res_num PK/FK,
changed_at PK,
changed_to,
changed_by
)

Step 4: Remove Unnecessary Tables

- No relation is a subset of another relation.
- No tables are removed.

Step 5: Check for Losslessness

Relationships:

1. Restaurant $\leftrightarrow$ Reservation
 - Shares rest-phone
 - lossless
2. Restaurant $\leftrightarrow$ ReservationSlot
 - shares rest-phone
 - lossless
3. Restaurant $\leftrightarrow$ Restaurant Hours
 - Shares rest-phone
 - lossless
4. User Account $\leftrightarrow$ Reservation
 - Shares email
 - lossless
5. User Account $\leftrightarrow$ Review
 - Shares email
 - lossless
6. Restaurant $\leftrightarrow$ Review
 - Shares rest-phone
 - lossless
7. Reservation $\leftrightarrow$ Reservation History
 - Shares res-num
 - lossless

3NF is lossless because every decomposition joins on a PK, and every child relation contains the parents key as a FK.

Final Relations from 3NF

Restaurant (

rest_phone PK,
rest_name,
street,
zip,
city,
state,
latitude,
price_range,
image_url,
rest_created

)

ReservationSlot (

rest_phone PK/FK,
slot_date PK,
slot_time PK,
slot_capacity

)

RestaurantHours (

rest_phone PK/FK
day-of-week PK
open-time,
close-time,
is_closed

)

UserAccount (

email PK,
pwd_hash,
user_role,
fname,
lname,
user_phone,
acct_created,
employer_phone FK

)

Reservation (

res_num PK
party_size,
special_req,
res_status,
res_created,
slot_date,
slot_time,
rest_phone FK,
email FK

)

Review (

email PK/FK,
rest_phone PK, FK,
rating,
comment,
review_created

)

ReservationHistory (

res_num PK/FK,
changed_at PK,
changed_to,
changed_by

)